

One prompt in. A governed, question-driven Power BI report out.

The real output of one dogfood run — no staging, no retouching. An HR dataset and a legacy Vietnamese analysis notebook were dropped into the inbox; the framework walked the full analytics lifecycle, a human holding every gate.

🔮 **The entire brief:** “We got an HR employee-performance dataset dropped in the inbox — I want a Power BI report out of it. Where do we start?”

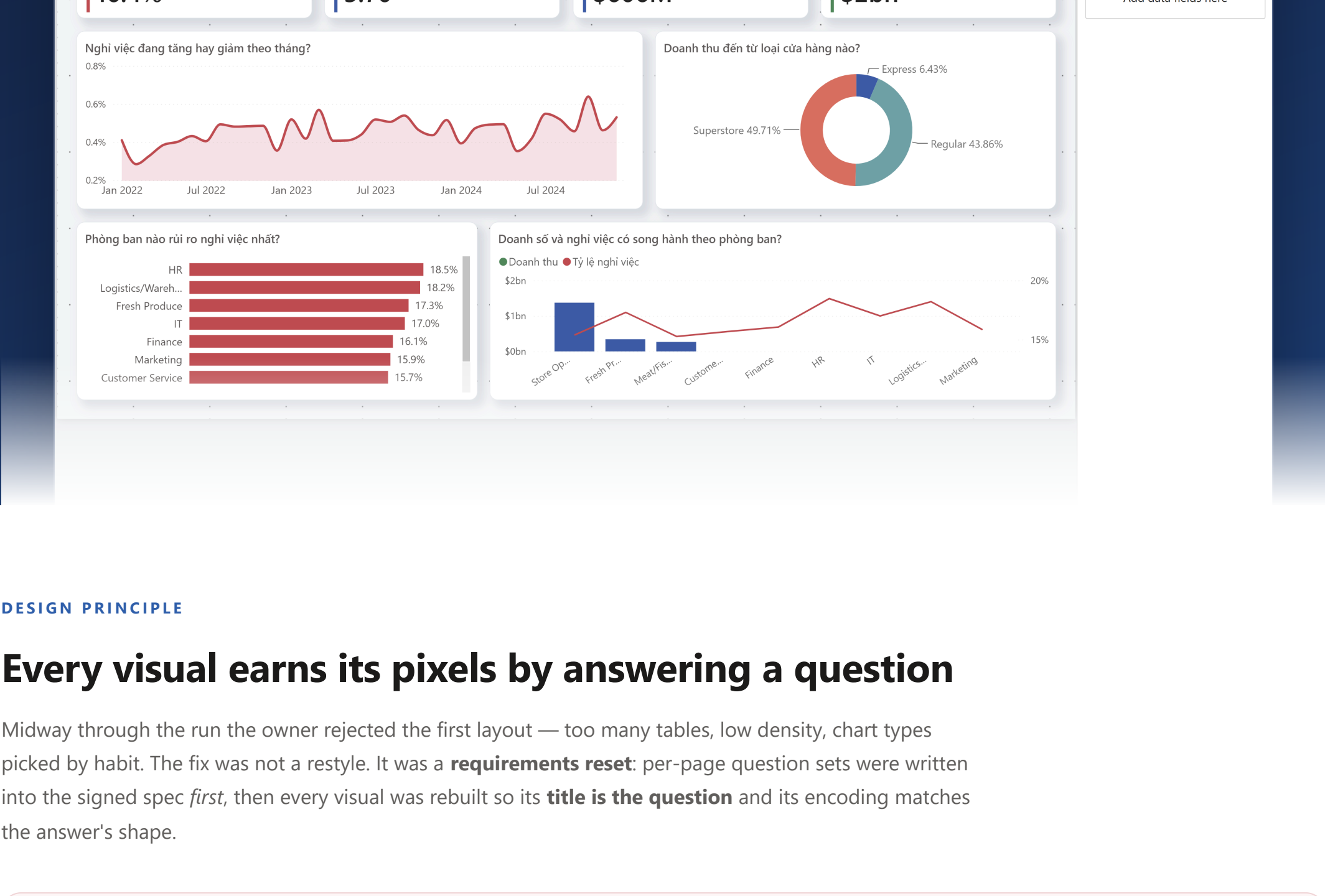
4
pages — every visual titled with the business question it answers

21
measures, all traced to governed metric cards

40
visuals - 7 encoding types - exactly ONE table report-wide

470k+
rows modeled (synthetic sample)

0.02pp
independent reproduction of the legacy attrition claim



DESIGN PRINCIPLE

Every visual earns its pixels by answering a question

Mixed by through the run the owner rejected the first layout — too many tables, low density, chart types picked by habit. The fix was not a restyle. It was a **requirements reset**: per-page question sets were written into the signed spec *first*, then every visual was rebuilt so its **title is the question** and its encoding matches the answer's shape.

BEFORE — v1

Column-driven design

“These fields exist, chart them.” The named junior failure mode.

- ✗ Titles were noun labels — “Attrition by Department”
- ✗ Two near-identical table+bar pages; bar/column everywhere
- ✗ 5 experience & quality columns profiled on day one, never used
- ✗ Density gaps the owner spotted on sight

AFTER — v2

Question-driven design

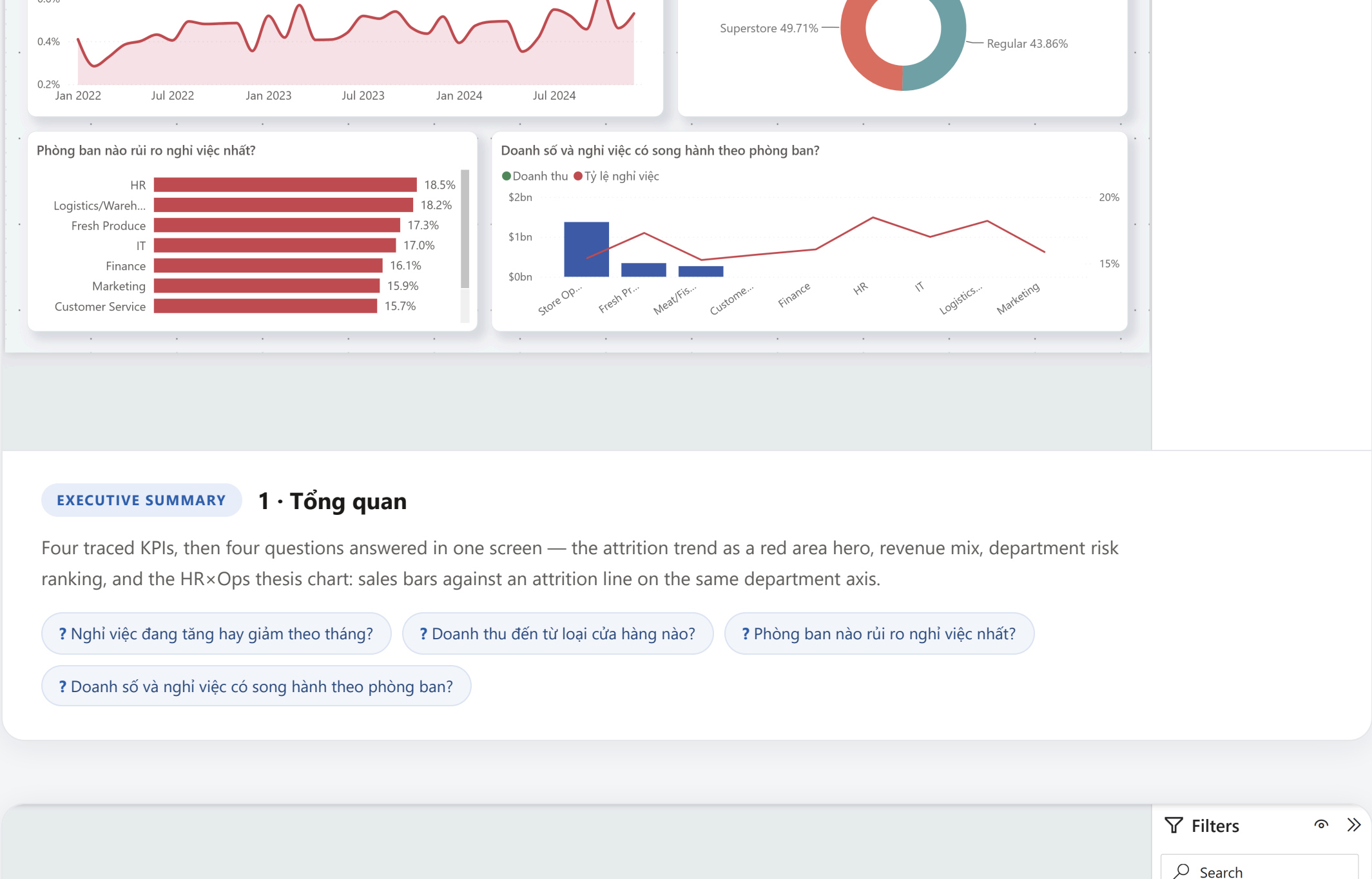
Requirements addendum signed before a single visual moved.

- ✓ Titles ARE the questions — “Phòng ban nào rủi ro nghỉ việc nhất?”
- ✓ 7 encodings by intent: area, donut, scatter x2, combo x2, stacked, ONE pivot
- ✓ 11 new driver measures, live-verified, under governed metric cards
- ✓ 40 visuals, layout gate green, render-verified page by page

THE REPORT

Four pages, one storyline

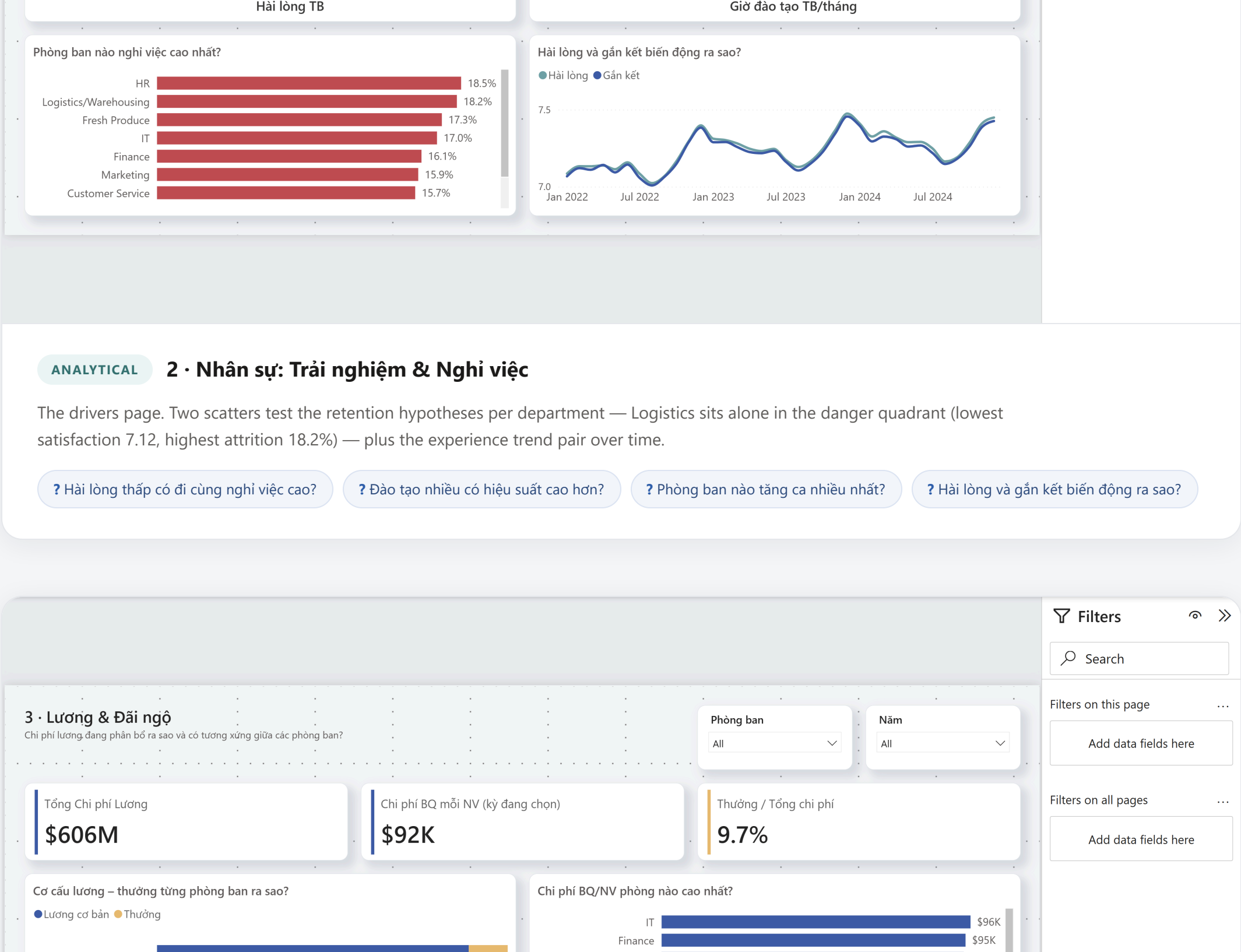
Overview → workforce drivers → compensation structure → store benchmark. Org theme on a 1280×720 twelve-column contract (machine-validated), Vietnamese questions for the exec audience, English identifiers underneath.



EXECUTIVE SUMMARY 1 - Tổng quan

Four traced KPIs; then four questions answered in one screen — the attrition trend as a red area hero, revenue mix, department risk ranking, and the HR×Ops thesis chart: sales bars against an attrition line on the same department axis.

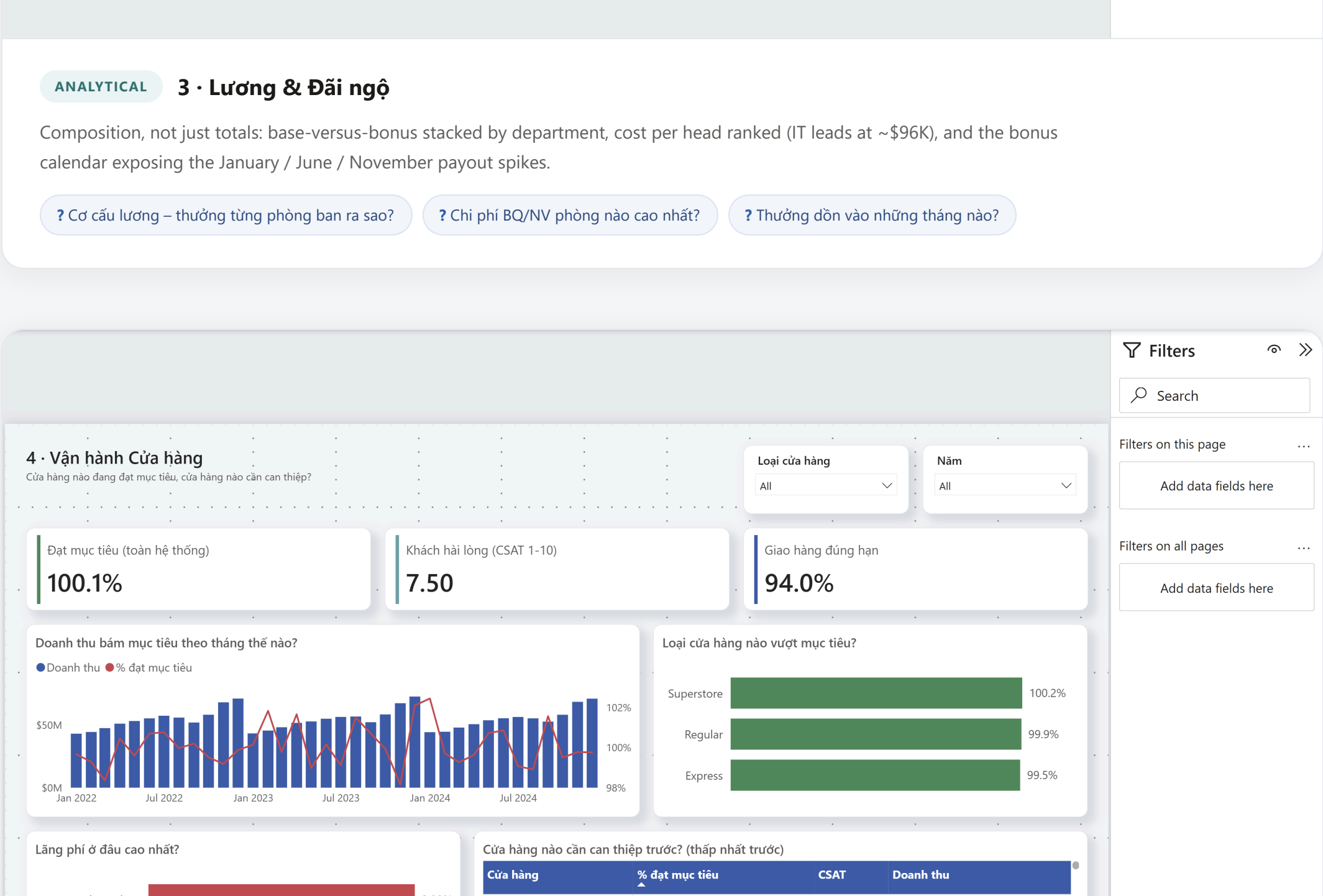
- ? Nghỉ việc đang tăng hay giảm theo tháng?
- ? Doanh thu đến từ loại cửa hàng nào?
- ? Phòng ban nào rủi ro nghỉ việc nhất?
- ? Doanh số và nghỉ việc có song hành theo phòng ban?



ANALYTICAL 2 - Nhân sự: Trải nghiệm & Nghỉ việc

The drivers page. Two scatters test the retention hypotheses per department — Logistics sits alone in the danger quadrant (lowest satisfaction 7.12, highest attrition 18.2%) — plus the experience trend pair over time.

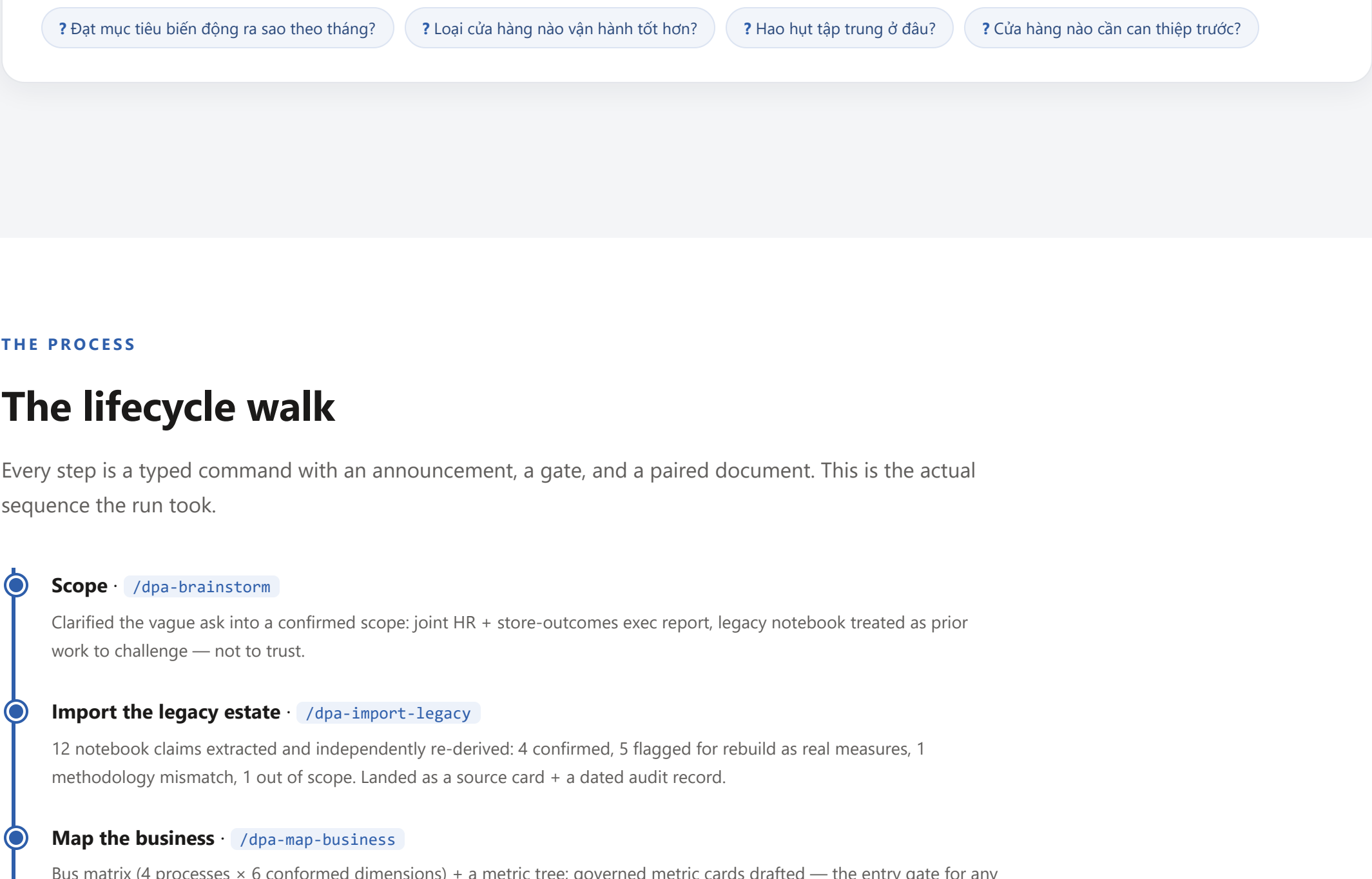
- ? Hải lương thấp có đi cùng nghỉ việc cao?
- ? Đào tạo nhiều có hiệu suất cao hơn?
- ? Phòng ban nào tăng ca nhiều nhất?
- ? Hải lương & gắn kết biến động ra sao?



ANALYTICAL 3 - Lương & Đào tạo

Composition, not just totals: base-versus-bonus stacked by department, cost per head ranked (IT leads at ~\$96K), and the bonus calendar exposing the January / June / November payout spikes.

- ? Cơ cấu lương - thưởng từng phòng ban ra sao?
- ? Chi phí BQ/NV phòng nào cao nhất?
- ? Thưởng dồn vào những tháng nào?



COMPARATIVE BENCHMARK 4 - Vận hành Cửa hàng

Target achievement by month and store type, waste hotspots, and the one table the whole report allows — the intervention shortlist, sorted worst-first with the service-quality context that explains each miss.

- ? Đạt mục tiêu biến động ra sao theo tháng?
- ? Loại cửa hàng nào vận hành tốt hơn?
- ? Hao hụt tập trung ở đâu?
- ? Cửa hàng nào cần can thiệp trước?

THE PROCESS

The lifecycle walk

Every step is a typed command with an announcement, a gate, and a paired document. This is the actual sequence the run took.

- Scope** · `//dpa-brainstorm`
Clarified the vague ask into a confirmed scope: joint HR + store-outcomes exec report, legacy notebook treated as prior work to challenge — not to trust.
- Import the legacy estate** · `//dpa-import-legacy`
12 notebook claims extracted and independently re-derived: 4 confirmed, 5 flagged for rebuild as real measures, 1 methodology mismatch, 1 out of scope. Landed as a source card + a dated audit record.
- Map the business** · `//dpa-map-business`
Bus matrix (4 processes × 6 conformed dimensions) + a metric tree; governed metric cards drafted — the entry gate for any measure that gets built.
- Lock requirements** · `//dpa-gather-requirements`
Intake brief: 5 user stories, 8 testable acceptance criteria, explicit out-of-scope. Signed before any build — and re-opened deliberately when the owner demanded the question-driven reset: spec addendum first, visuals second.
- Build the model** · `//dpa-build-model`
Star schema (6 tables, 7 relationships, 21 measures) in TMDL — typed source contract from the safe profiles, explicit casts and date culture, then the **DAX smoke gate** against a live engine before any report work.
- Build the report** · `//dpa-build-report`
Locked design contract (per-page question sets + layout_contract) → generator-authored PBIR → three validators → Desktop render-verify, every page, every visual.
- Verify & document** · `//dpa-check`
Evidence record for the smoke gate + render review; report card, model card, dictionary, lineage all regenerated and validated. No citation, no claim.

THE HONEST PART

What the framework caught

Real defects produced during the build that static generation alone would have shipped. Each one is now a permanent gate, so the next run cannot repeat it.

A flat-line KPI that looked plausible

Attrition rendered —19.9% on every single month —the measure silently ignored the date filter (no relationship between the employee dim and the date table). **Now gated:** the Stage-4 DAX smoke gate fails any measure that is constant across all slice values.

A window that only counted day-1 events

The first fix windowed exits on month-start dates — monthly attrition rendered ~10x too low. Caught by reading the rendered chart against source-derived expectations. **Now gated:** the window-edge probe reconciles single-month against year totals.

Numbers that arrived as text

SUM cannot work with values of type STRING — Excel columns imported without explicit casts; DD/MM/YYYY dates mis-parsed by the default locale (4,873 of 7,500 rows errored). **Now gated:** the typed source contract — every gateway casts every column, locale-ambiguous dates carry a culture.

Four ways to make Desktop refuse the file

A near-miss: `schema URL`, a missing `definition.pbix`, a relative `FileA.Contents` path, a dataset-only `pbix` — each found the hard way, each now a deterministic check (PBIR-008..011) that runs before the first Desktop open.

A layout the owner rejected on sight

Table-heavy pages, low chart density, encodings picked by habit — the root cause was a requirements gap, not a styling gap. **Now framework law:** per-page question sets in the spec before any visual is placed; visual titles are the questions (report-design pack §2b rhythm rules).

Blank rows nobody should ship

A dimension dropped into a table's Values well rendered phantom blank rows. **Now protocol:** every page, every visual is render-reviewed against a live Desktop before sign-off — screenshots are the evidence.

And the payoff: after the rebuild — never copying the legacy numbers — the framework's independently derived 2024 attrition (5.77%) reproduced the original analyst's claim (~5.79%) to within 0.02 percentage points. Two independent methods, one answer: that is what “verified” means here.

BEYOND THE PIXELS

The governed exhaust

A report is the visible 20%. The other 80% is what makes the next engagement faster and the numbers defensible.

Knowledge cards

- `knowledge/sources/hr-employee-performance.pbix` — provenance, grain, contract
- `knowledge/business-model.md` — bus matrix + metric tree with drivers
- `knowledge/metrics/hr/*` + `sales/*` — 6 governed metric cards
- `knowledge/models/hr-performance/` — model card + generated dictionary
- `knowledge/reports/hr-performance.md` — report card, verified axis

Dated records

- Claim-challenge audit — 12 legacy claims adjudicated
- DAX smoke-gate evidence — live-engine probe results
- Decision records DEC-0029/0030 — what was ruled and why

Machine gates added

- PBIR-008..011 openability checks (+7 tests)
- Stage-4 entry/exit contracts (typed sources, DAX smoke gate)
- Report-design rhythm rules R1–R7 + the question-first spec gate

Artifacts in git

- `Fabric/HR-Performance.SemanticModel/` — TMDL, canonical
- `Fabric/HR-Performance.Report/` — PBIR, 40 visuals
- Locked spec + signed v2 question-set addendum in `_brief/`

All data on this page is synthetic (Datapot Sample Retail Co, 7,500 employees / 150 stores, 2022-2024). Every aggregate shown is owner-approved per the framework's data-handling rules.

Datapot Analytics Framework — an AI teammate for the Power BI / Fabric lifecycle, with a human at every gate.

Start here: `scope` → type `map` in your agent tool. The agent contract lives in `contract`